Solution: Longest Subarray with Ones after Replacement

Problem Statement

Given an array containing 0s and 1s, if you are allowed to **replace no more than 'k' 0s with 1s**, find the length of the **longest contiguous subarray having all 1s**.

Example 1:

```
Input: Array=[0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1], k=2
Output: 6
Explanation: Replace the '0' at index 5 and 8 to have the longest contig
uous subarray of 1s having length 6.
```

Example 2:

```
Input: Array=[0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 1, 1], k=3
Output: 9
Explanation: Replace the '0' at index 6, 9, and 10 to have the longest c
ontiguous subarray of 1s having length 9.
```

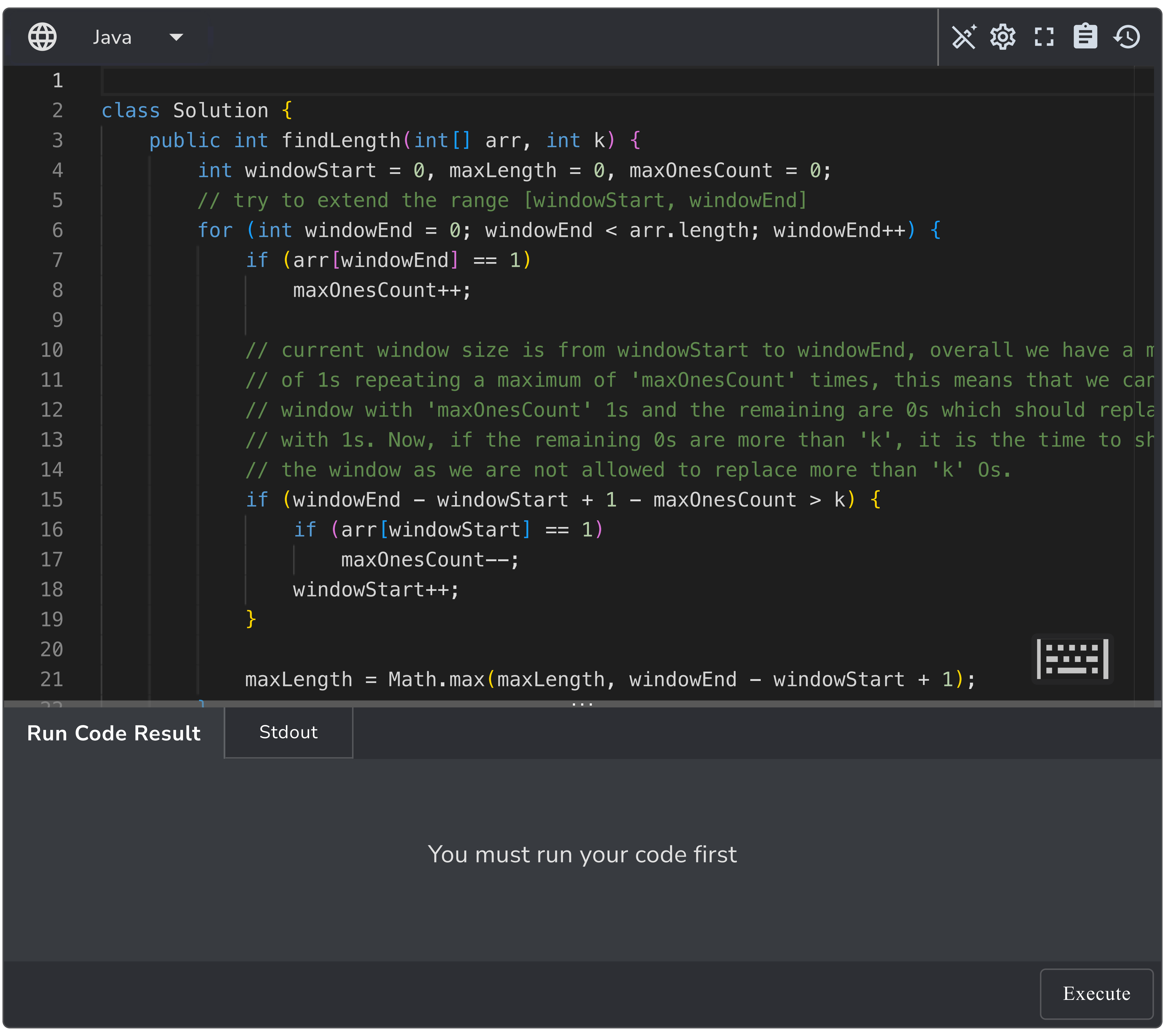
Solution

This problem follows the **Sliding Window** pattern and is quite similar to Longest Substring with same Letters after Replacement. The only difference is that, in the problem, we only have two characters (1s and 0s) in the input arrays.

Following a similar approach, we'll iterate through the array to add one number at a time in the window. We'll also keep track of the maximum number of repeating 1s in the current window (let's call it `maxOnesCount`). So at any time, we know that we can have a window with 1s repeating `maxOnesCount` time, so we should try to replace the remaining 0s. If we have more than 'k' remaining 0s, we should shrink the window as we are not allowed to replace more than 'k' 0s.

Code

Here is what our algorithm will look like:



```java

class Solution {
    public int findLength(int[] arr, int k) {
        int windowStart = 0, maxLength = 0, maxOnesCount = 0;
        // try to extend the range [windowStart, windowEnd]
        for (int windowEnd = 0; windowEnd < arr.length; windowEnd++) {
            if (arr[windowEnd] == 1)
                maxOnesCount++;

            // current window size is from windowStart to windowEnd, overall we have a m
            // of 1s repeating a maximum of 'maxOnesCount' times, this means that we car
            // window with 'maxOnesCount' 1s and the remaining are 0s which should repla
            // with 1s. Now, if the remaining 0s are more than 'k', it is the time to sh
            // the window as we are not allowed to replace more than 'k' Os.
            if (windowEnd - windowStart + 1 - maxOnesCount > k) {
                if (arr[windowStart] == 1)
                    maxOnesCount--;
                windowStart++;
            }

            maxLength = Math.max(maxLength, windowEnd - windowStart + 1);
```

Time Complexity

The above algorithm's time complexity will be $O(N)$, where 'N' is the count of numbers in the input array.

Space Complexity

The algorithm runs in constant space $O(1)$.